FACULTY OF ENGINEERING AND TECHNOLOGY (FET)

DEPARTMENT OF COMPUTER ENGINEERING

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

DESIGN AND IMPLEMENTATION OF A MOBILE APPLICATION FOR COLLECTION OF USER EXPERIENCE DATA FROM MOBILE NETWORK SUBSCRIBERS

Submitted By Group 2

S/N	NAME	MATRICULE
1.	EFUETNGONG GENESIS TAZISONG	FE23A041
2.	MATUTE MOLUA FRITZ	FE23A087
3.	NWANTOLY STEPHILL FAVOUR BENYELLA	FE23A127
4.	NYANYOH DIVINE-FORTUNE BANFILA	FE23A128
5.	TENKEN MANPLA ULRICH	FE23A170

COURSE INSTRUCTOR: Dr. Valery Nkemeni

CEF440: INTERNET PROGRAMMING AND MOBILE PROGRAMMING

2025 / 2026 Academic Year

Date Submitted: 11/05/2026

This document contains the Software Requirements Specification (SRS) for the proposed Mobile Network User Experience Data Collection System. The information contained herein is intended strictly for academic and project evaluation purposes.

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Contents

SOFTWARE REQUIREMENT SPECIFICATION (SRS)

1. Introduction

Mobile communication has become an essential part of everyday life. As mobile networks continue to expand and evolve, users increasingly expect reliable connectivity, fast internet speeds, stable calls, and excellent overall service quality. However, many mobile subscribers still experience poor network performance, call drops, slow internet access, latency issues, and unstable connections.

Traditional methods used by mobile network operators to monitor network performance primarily focus on technical network parameters and often fail to adequately capture the actual experiences of end users. This creates a gap between network statistics and the real Quality of Experience (QoE) perceived by subscribers.

This project proposes the design and implementation of a mobile application capable of collecting both subjective and objective Quality of Experience data from mobile network subscribers in real time. The application will gather user feedback together with network performance metrics such as signal strength, network type, latency, jitter, bandwidth, packet loss, timestamp, and location information.

The collected information will assist network operators in identifying service problems, improving network optimization, enhancing customer satisfaction, and supporting data-driven decision-making.

2. Purpose of the System

The purpose of the proposed system is to provide an efficient, automated, and user-friendly platform for collecting and analyzing mobile network user experience data.

The system is intended to:

- Collect real-time user feedback regarding network quality
- Monitor mobile network performance automatically
- Store and analyze Quality of Experience data
- Generate reports and statistical visualizations
- Help mobile network operators improve service quality
- Support better customer experience management
- Enable proactive identification of network problems

3. Scope of the Project

The scope of this project covers the design and implementation of an Android-based mobile application capable of:

- Running continuously in the background
- Collecting network performance parameters
- Gathering user satisfaction feedback
- Recording timestamps and location information
- Storing collected data in a database

- Generating charts and analytical reports
- Supporting Quality of Experience evaluation

The project mainly targets mobile network subscribers and network operators within Cameroon.

The system focuses primarily on Android devices because of their widespread use among mobile subscribers.

4. Objectives of the System

The major objectives of the system are:

Main Objective

To design and implement a mobile application that enables real-time collection and analysis of user experience data from mobile network subscribers.

Specific Objectives

The system shall:

- Collect user satisfaction ratings
- Monitor network performance automatically
- Record network-related parameters
- Store data securely in a database
- Generate statistical reports and visual charts
- Improve Quality of Experience analysis
- Assist mobile network operators in network optimization
- Provide a user-friendly interface for subscribers

5. Stakeholders of the System

5.1 Mobile Network Subscribers (Users)

These are the primary users of the application. They interact directly with the system by providing feedback concerning their mobile network experiences.

Users help the system by:

- Rating network quality
- Reporting call drops
- Evaluating internet performance
- Sharing Quality of Experience information

Their feedback enables accurate assessment of network service quality.

5.2 Mobile Network Operators

Examples include:

- ➢ MTN Cameroon

- Orange Cameroon
- Camtel

These organizations use the collected data to:

- Improve network performance
- Detect service issues
- Optimize network coverage
- Reduce customer complaints
- Improve customer satisfaction
- Support decision-making processes

5.3 System Developers

Developers are responsible for:

- Designing the application
- Implementing system functionalities
- Maintaining the application
- Managing updates and bug fixes
- Ensuring system reliability and security

5.4 Regulatory Authorities

Regulatory bodies ensure that the system complies with:

- Telecommunications regulations
- Data privacy policies
- Security standards
- Ethical data collection practices

6. Overall Description

The proposed system is a mobile application that combines active user feedback with passive network monitoring techniques.

The application operates in the background and periodically prompts users to provide feedback regarding their network experiences.

At the same time, the application automatically collects network and device parameters such as:

- Signal strength
- Network type
- Internet speed
- Packet loss
- Latency
- Jitter
- Device information
- GPS location

- Timestamp

The collected data is stored in a centralized database where it can later be analyzed and visualized using charts and reports.

7. System Features

The proposed system includes the following major features:

7.1 User Feedback Collection

The application allows users to:

- Rate network quality
- Report call drops
- Evaluate internet speed
- Assess service reliability

7.2 Automatic Network Monitoring

The system automatically collects:

- Signal strength
- Network type
- Latency
- Packet loss
- Internet speed
- Device details

7.3 Background Operation

The application runs continuously in the background with minimal interruption to normal device usage.

7.4 Data Storage

Collected data is securely stored in a database for future analysis.

7.5 Data Visualization

The system generates:

- Charts
- Pivot tables
- Graphs
- Statistical summaries

7.6 Report Generation

The system supports report generation for:

- Network analysis
- QoE evaluation

- Performance monitoring

8. Functional Requirements

Functional requirements describe what the system must do.

8.1 User Registration and Identification

The system shall:

- Allow user identification
- Store basic device information
- Associate collected data with users anonymously where necessary

8.2 Feedback Collection

The system shall:

- Prompt users for feedback periodically
- Allow users to rate network quality
- Record satisfaction levels
- Capture call drop experiences

8.3 Network Monitoring

The application shall automatically monitor:

- Signal strength
- Network type
- Internet speed
- Latency
- Packet loss
- Jitter

8.4 Location Tracking

The system shall:

- Record user location during data collection
- Associate QoE data with geographical locations

8.5 Data Storage

The system shall:

- Store collected information securely
- Maintain timestamps for all records
- Organize data into structured tables

8.6 Report Generation

The system shall:

- Generate reports automatically
- Produce statistical summaries

- Generate charts and visual analytics

8.7 Background Services

The application shall:

- Operate continuously in the background
- Perform scheduled monitoring tasks
- Minimize resource consumption

9. Non-Functional Requirements

Non-functional requirements describe the quality attributes of the system.

9.1 Performance Requirements

The system should:

- Operate efficiently in real time
- Minimize battery consumption
- Minimize internet data usage
- Respond quickly to user interactions

9.2 Security Requirements

The system should:

- Protect user privacy
- Encrypt sensitive data
- Use secure communication protocols
- Prevent unauthorized access

9.3 Reliability Requirements

The system should:

- Operate continuously with minimal failures
- Prevent data loss
- Recover gracefully from errors

9.4 Usability Requirements

The system should:

- Provide a simple user interface
- Be easy to navigate
- Require minimal technical knowledge

9.5 Compatibility Requirements

The application should:

- Support Android devices

- Operate on multiple Android versions
- Support different screen sizes

10. System Architecture Overview

The system architecture consists of the following major components:

10.1 Mobile Application Layer

Responsible for:

- User interaction
- Data collection
- Background monitoring
- Device communication

10.2 Database Layer

Responsible for:

- Data storage
- Data retrieval
- Data organization
- Data management

10.3 Analytics Layer

Responsible for:

- Data processing
- QoE analysis
- Chart generation
- Statistical reporting

10.4 Administration Layer

Responsible for:

- Monitoring system performance
- Managing collected data
- Generating reports

11. Data Requirements

The system shall collect and store the following data:

Data Type	Description
User Feedback	User satisfaction ratings
Signal Strength	Network signal measurements
Network Type	2G, 3G, 4G, 5G
Location Data	GPS coordinates
Device Information	Device model and OS
Timestamp	Date and time of collection
Internet Performance	Speed, latency, jitter

12. User Interface Requirements

The user interface should:

- Be simple and attractive
- Support easy navigation
- Display feedback forms clearly
- Provide readable charts and reports
- Be responsive across devices

The interface should also:

- Use clear labels
- Minimize user complexity
- Support accessibility and usability

13. Security Requirements

To ensure data privacy and confidentiality, the system shall implement:

- User permission management
- Secure authentication mechanisms
- Data encryption
- Secure database access
- Secure network communication

The application must comply with data protection and privacy regulations.

14. Performance Requirements

The application should:

- Run efficiently in the background
- Avoid excessive CPU usage
- Consume minimal battery power
- Handle large volumes of collected data
- Upload data efficiently

The system should maintain acceptable performance under normal operating conditions.

15. Software and Hardware Requirements

15.1 Software Requirements

Development tools may include:

- Android Studio
- Java/Kotlin
- Firebase/MySQL
- Microsoft Excel
- Google Forms

15.2 Hardware Requirements

The system requires:

- Android smartphone
- Internet connection
- GPS-enabled device
- Server or cloud storage infrastructure

16. Constraints and Limitations

Several limitations may affect the system:

- Dependence on internet connectivity
- Device compatibility limitations
- Battery consumption concerns
- GPS accuracy variations
- User reluctance toward data sharing

These constraints must be considered during implementation.

17. Assumptions and Dependencies

The system assumes that:

- Users possess Android smartphones
- Users grant necessary permissions
- Internet connectivity is available
- GPS services are enabled
- Mobile devices support background services

The system also depends on:

- Android APIs
- Database availability
- Cloud synchronization services

18. Use Case Descriptions

Use Case 1: Submit User Feedback

Actor

Mobile Subscriber

Description

The user provides ratings regarding network quality.

Flow

1. User opens the application
2. Feedback form is displayed
3. User submits ratings
4. System stores feedback

Use Case 2: Automatic Network Monitoring

Actor

System

Description

The system automatically collects network parameters.

Flow

1. Background service starts
2. System collects network data
3. Data is timestamped
4. Data is stored in database

Use Case 3: Generate Reports

Actor

Administrator/Operator

Description

The system generates charts and reports from collected data.

Flow

1. Administrator requests analysis
2. System processes stored data
3. Charts and reports are generated

19. Future Enhancements

Future improvements may include:

- iOS support
- AI-based network prediction
- Real-time alerts
- Advanced analytics dashboards
- Machine learning integration
- Offline data synchronization
- Automated QoE recommendations

20. Conclusion

This Software Requirement Specification (SRS) document presents the detailed requirements for the design and implementation of a mobile application for collecting user experience data from mobile network subscribers.

The document defines the system objectives, functional requirements, non-functional requirements, system architecture, stakeholder expectations, and operational constraints.

The proposed system will enable efficient collection and analysis of Quality of Experience data, helping mobile network operators improve service quality, optimize network performance, and enhance customer satisfaction.

The SRS serves as a professional guide for system developers, designers, testers, and stakeholders throughout the software development life cycle.